

Plotting the Dark Matter Distribution of a Quasar-Galaxy Lensing System

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I. Introduction

My project is focused on plotting the distribution of dark matter of the galaxy in the gravitational lensing system, DES J0408-5354. The main galaxy is bending the light of a distant quasar behind it, creating three distinct lensing images and a clear Einstein radius arching around the galaxy. To do this, I will utilize data about the system to find the mass of the dark matter enclosed within the Einstein radius. I will then fit a mass-density model to find the scale radius and scale density which I will then use to plot the density as a function of radius from the center of the galaxy to visualize the distribution. To confirm my model is properly fitting the data, I have generated mock test data.

II. Data

I have compiled the necessary data about the DES system from various papers studying its properties (Shajib et al., 2020), (Lin et al., 2017), (Agnello et al., 2017).

PROPERTY	VALUE
Einstein radius (arcs)	1.75
Redshift of lens	0.597
Redshift of source	2.375
Stellar mass ($\log(\frac{M^*}{M_\odot})$)	11.49

III. Finding Dark Matter Mass

To determine the mass of dark matter, I started by calculating the total enclosed mass within the Einstein radius of the galaxy, including the mass of both dark and stellar matter. To do this, I used the equation for the Einstein radius but rearranged to solve for mass:

$$M_E = \frac{c^2}{4G} \frac{D_S D_L}{D_{LS}} \theta^2$$

Where D_S , D_L , and D_{LS} are the distance to the source, the distance to the lens, and the distance between the source and the lens respectively. I found these using the `angular_diameter_distance` function of the `astropy.cosmology` package. G and c are physical constants, and θ is the Einstein radius in radians.

I then took the log of my calculated mass and subtracted the stellar mass from it to find the mass of dark matter enclosed within the Einstein radius.

IV. The Model

For this project, I used the Navarro-Frenk-White profile, which is the most commonly used profile to model the spatial mass and distribution of dark matter halos. It approximates dark matter halos as having very dense centers that get less dense further from the center.

I started by fitting the mass profile to my data to find a mass and then minimized the square of the distance between my modeled mass and my calculated mass to find the parameters.

$$M = 4\pi\rho_s R_s^3 \left[\ln\left(\frac{R_s + R}{R_s}\right) - \frac{R}{R_s + R} \right]$$

The parameters I minimized were the scale density, ρ_s and scale radius, R_s . I also used test data that I came up with to determine if my model fit and objective function were properly finding the best parameters.

V. Visualization

After finding the parameters, I used the density profile of NFW to plot density as a function of radius away from the center of the galaxy and got a plot that generally resembles most NFW graphs.

